

Discovering Hidden Structure in Big Data with Simple Math

Dr. Jishan Ahmed

Data Science Assistant Professor

Department of Mathematics



Programs

- ☐ Master of Science in Data Science
- ☐ Computational Statistics & Data Science
- ☐ Data Science & Applied Statistics Certificate of Proficiency
- ☐ Mathematics
- ☐ Applied Mathematics
- ☐ Teaching



Linear Algebra is all you Need for Machine Learning

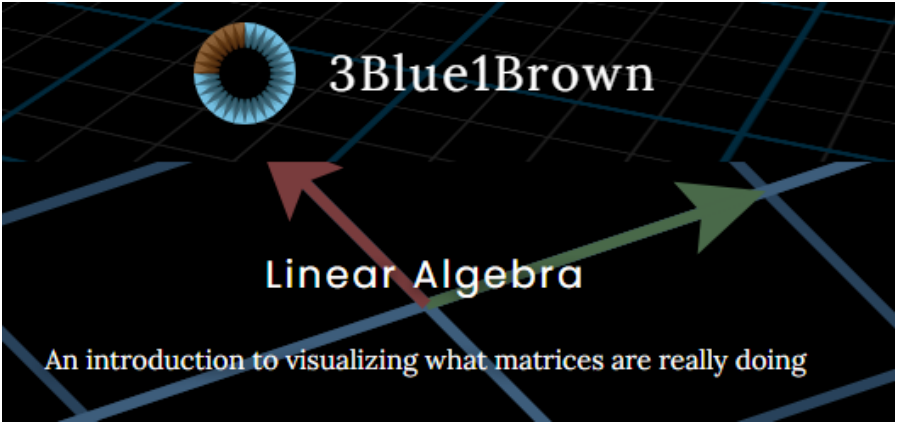
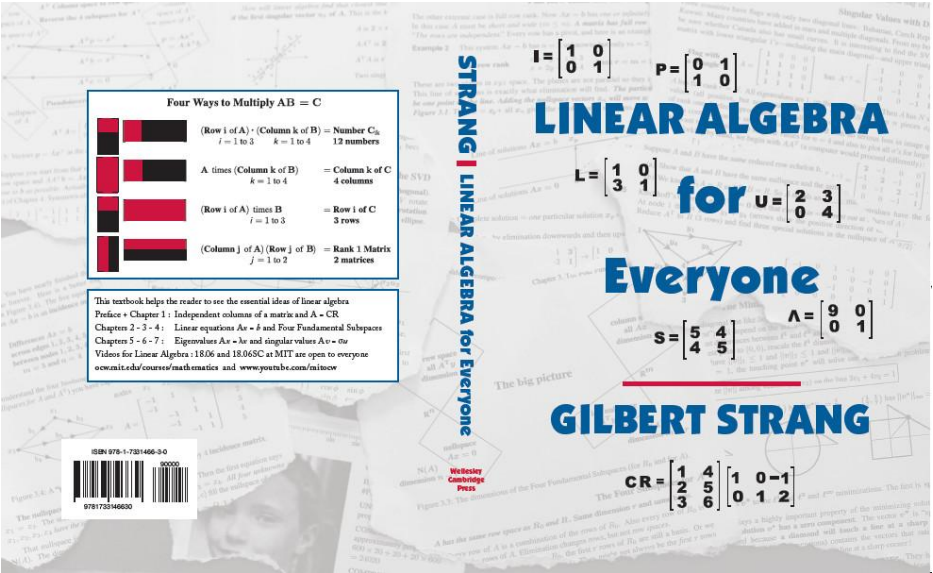


Image Source: <https://math.mit.edu/~gs/>

Representing Images

What a human sees!



Representing Images

What a computer sees!

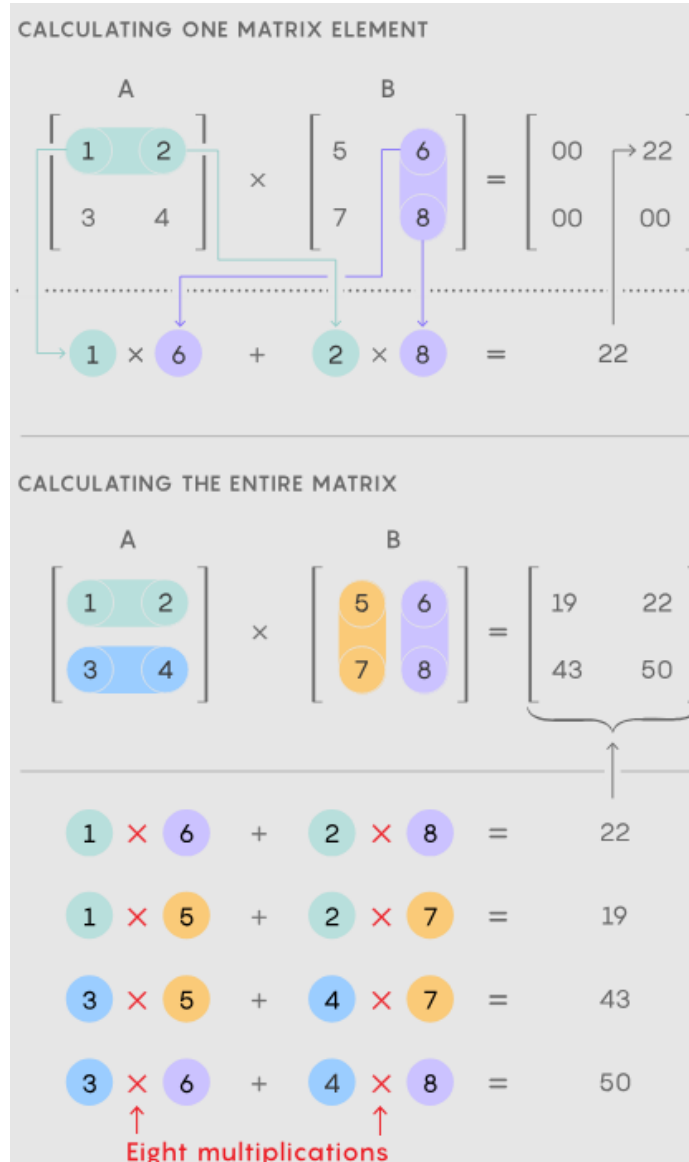
Annotated Pixel Values

160	155	143	154	176	158	123	114	131	136
147	161	161	155	118	112	141	161	155	148
151	148	127	99	149		143	134	135	147
148	124	143	177	147	117	146	148	147	128
103	163		107	150	164	140	115	112	104
151	106	79	124	131	121	91	93	112	147
74	86	118	114	90	68	106	111	125	72
111	103	78	93	66	107	103	73	80	83
83	89	104	64	99	72	46	76	76	72
71	98	69	71	63	56	82	64	68	98

What a human sees!



Matrix Multiplication



<http://matrixmultiplication.xyz/>

Image Source: <https://www.quantamagazine.org/>
and <https://collider.com/>



WEBER STATE UNIVERSITY

Matrix Multiplication

$$\begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \end{bmatrix} \times \begin{matrix} B \\ A \end{matrix} = \begin{matrix} BA \text{ (Horizontally Flipped)} \end{matrix}$$


Identity Matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transpose of a Matrix

transpose of a matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}_{2 \times 3} \quad A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}_{3 \times 2}$$

Matrix A (Original Image)



A^T (Transpose of A)



$(A^T)^T$ (Transpose of A^T)



Singular Value Decomposition

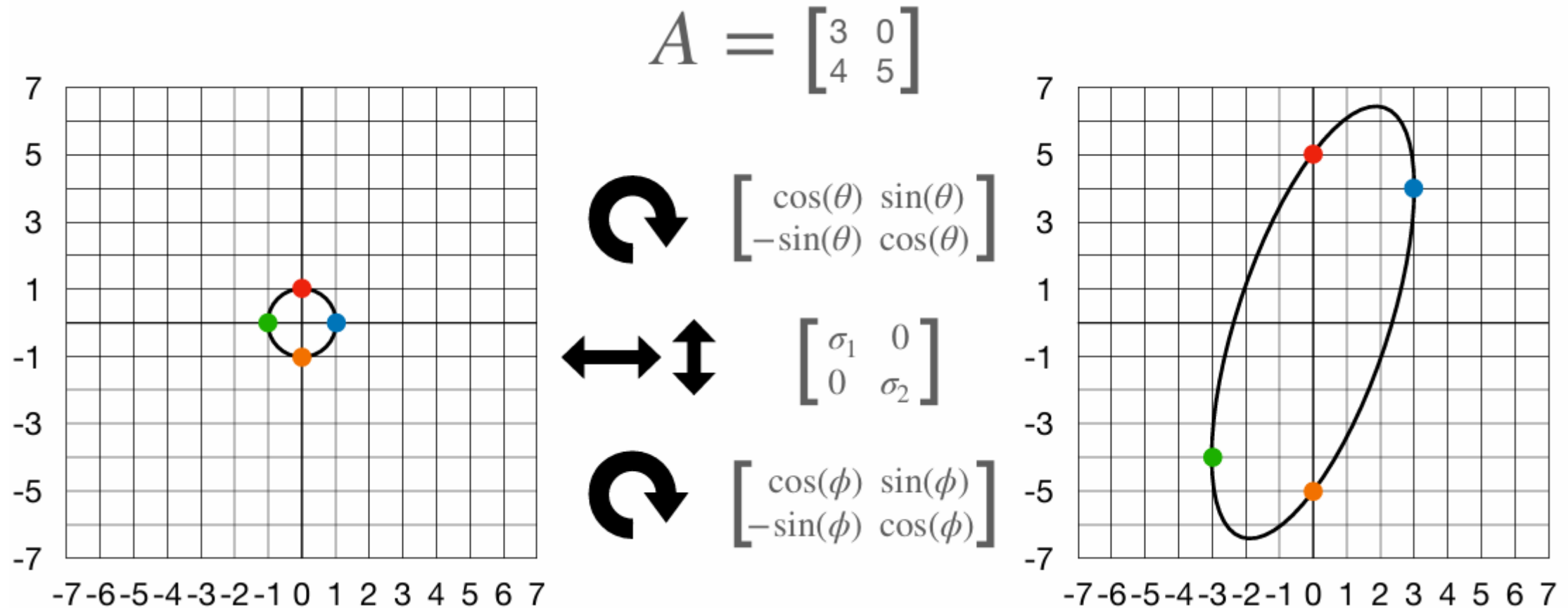


Image Source: Luis Serrano



Singular Value Decomposition

FROM THE MAKERS OF WOLFRAM LANGUAGE AND MATHEMATICA

 **WolframAlpha**

singular value decomposition [[3,0],[4,5]]

 NATURAL LANGUAGE  MATH INPUT

 EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Input

singular value decomposition $\begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$

Result Exact forms

$M = U \cdot \Sigma \cdot V^\dagger$

where

$M = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$

$U = \begin{pmatrix} 0.316228 & -0.948683 \\ 0.948683 & 0.316228 \end{pmatrix}$

$\Sigma = \begin{pmatrix} 6.7082 & 0 \\ 0 & 2.23607 \end{pmatrix}$

$V = \begin{pmatrix} 0.707107 & -0.707107 \\ 0.707107 & 0.707107 \end{pmatrix}$

$m^\dagger$ gives the conjugate transpose of m

POWERED BY THE WOLFRAM LANGUAGE



Singular Value Decomposition

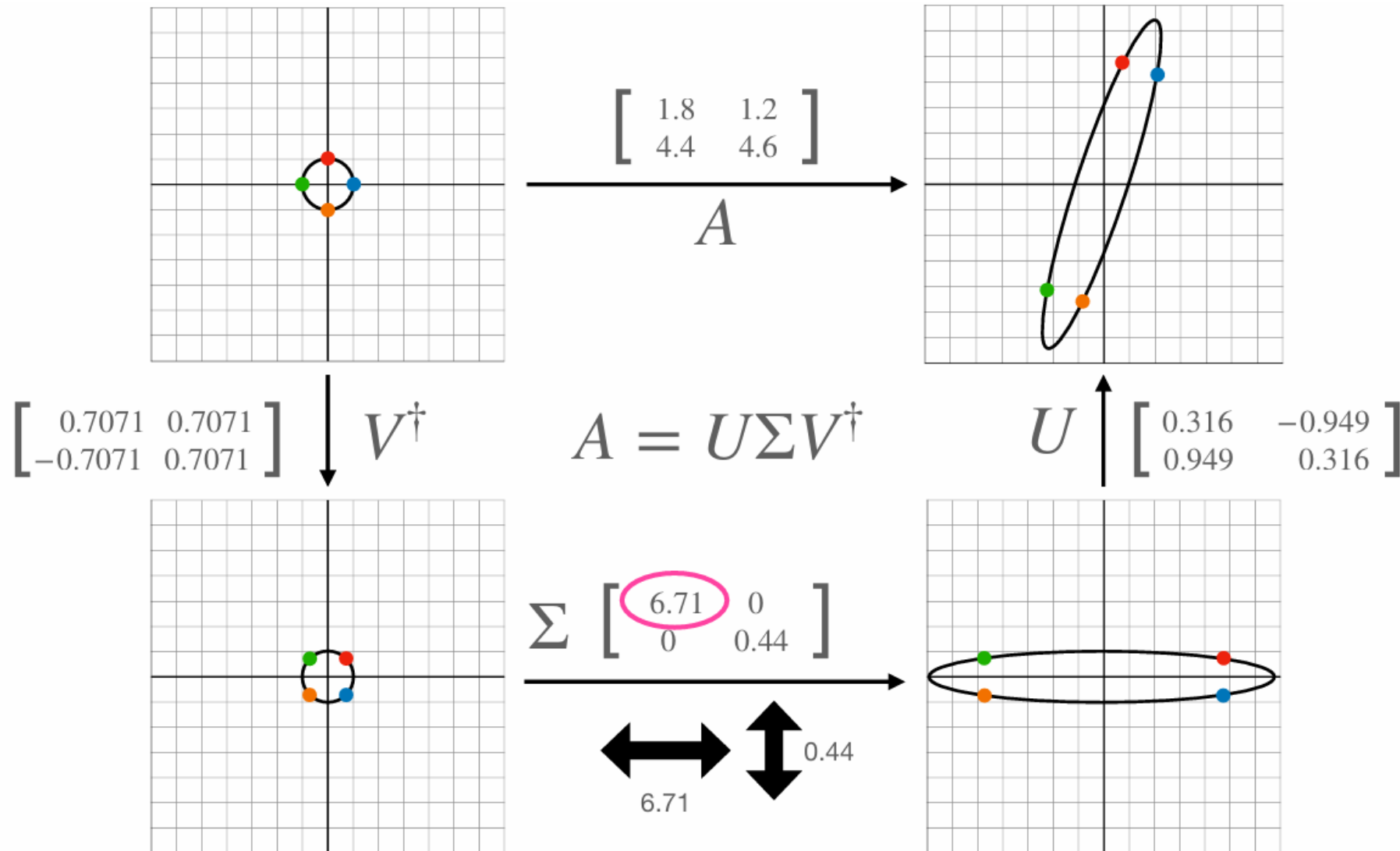


Image Source: Luis Serrano



Singular Value Decomposition

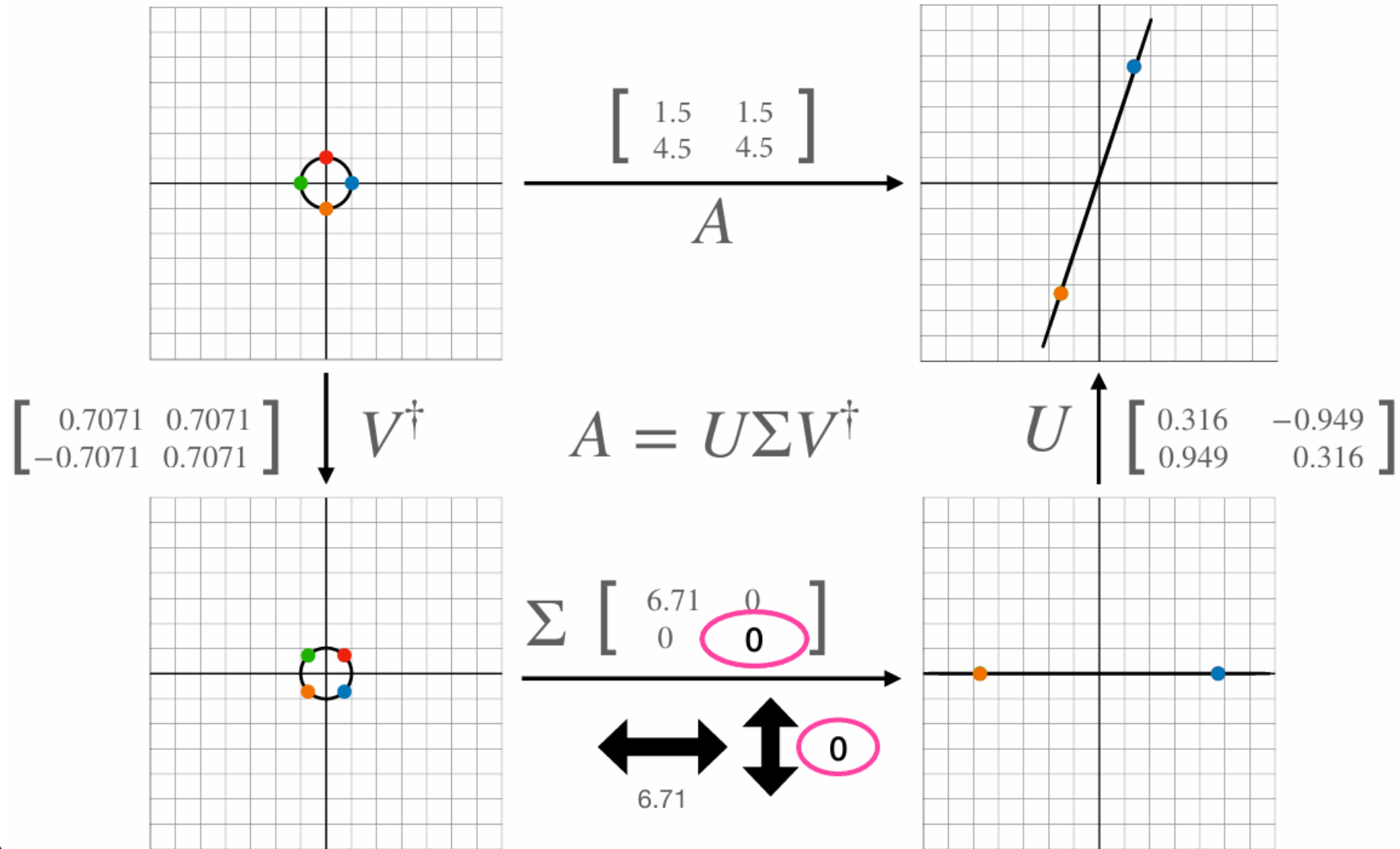


Image Source: Luis Serrano



Singular Value Decomposition

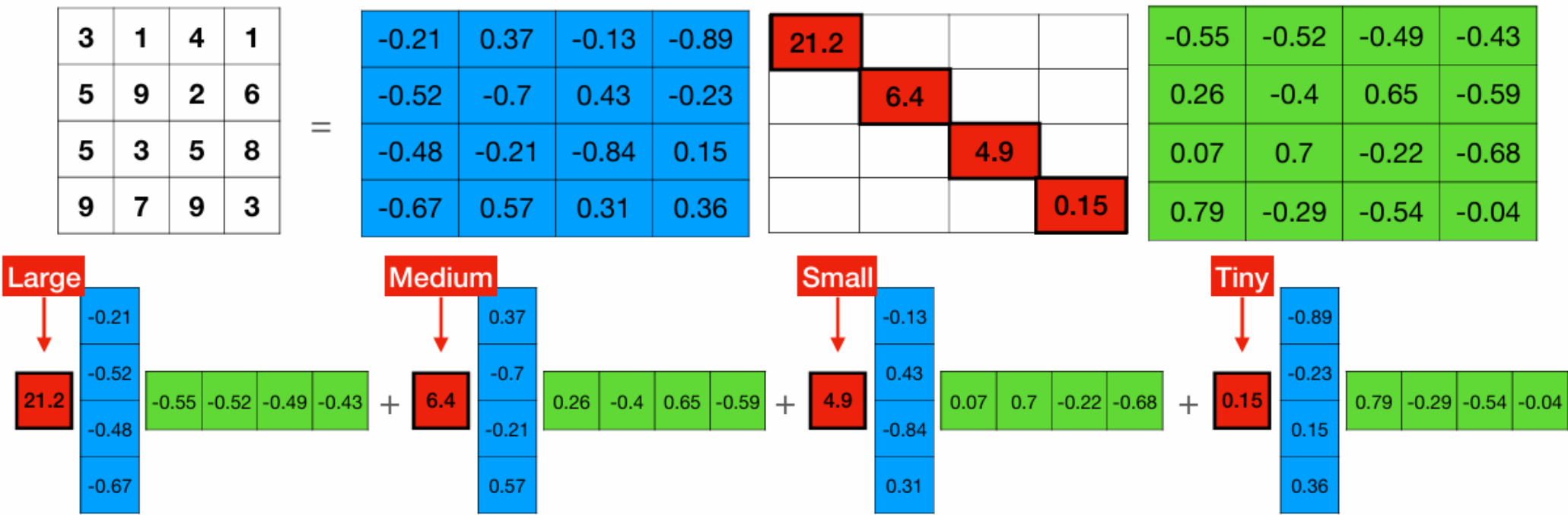
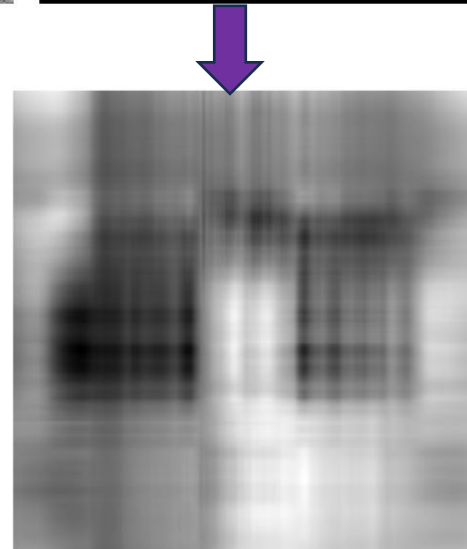
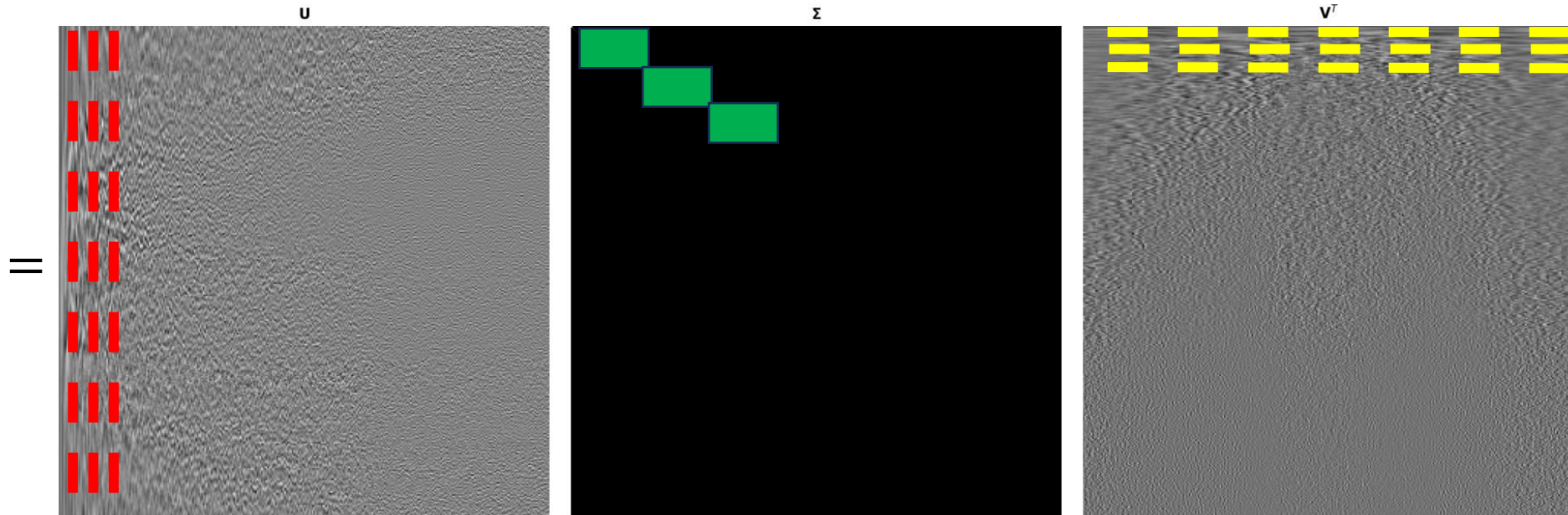


Image Source: Luis Serrano

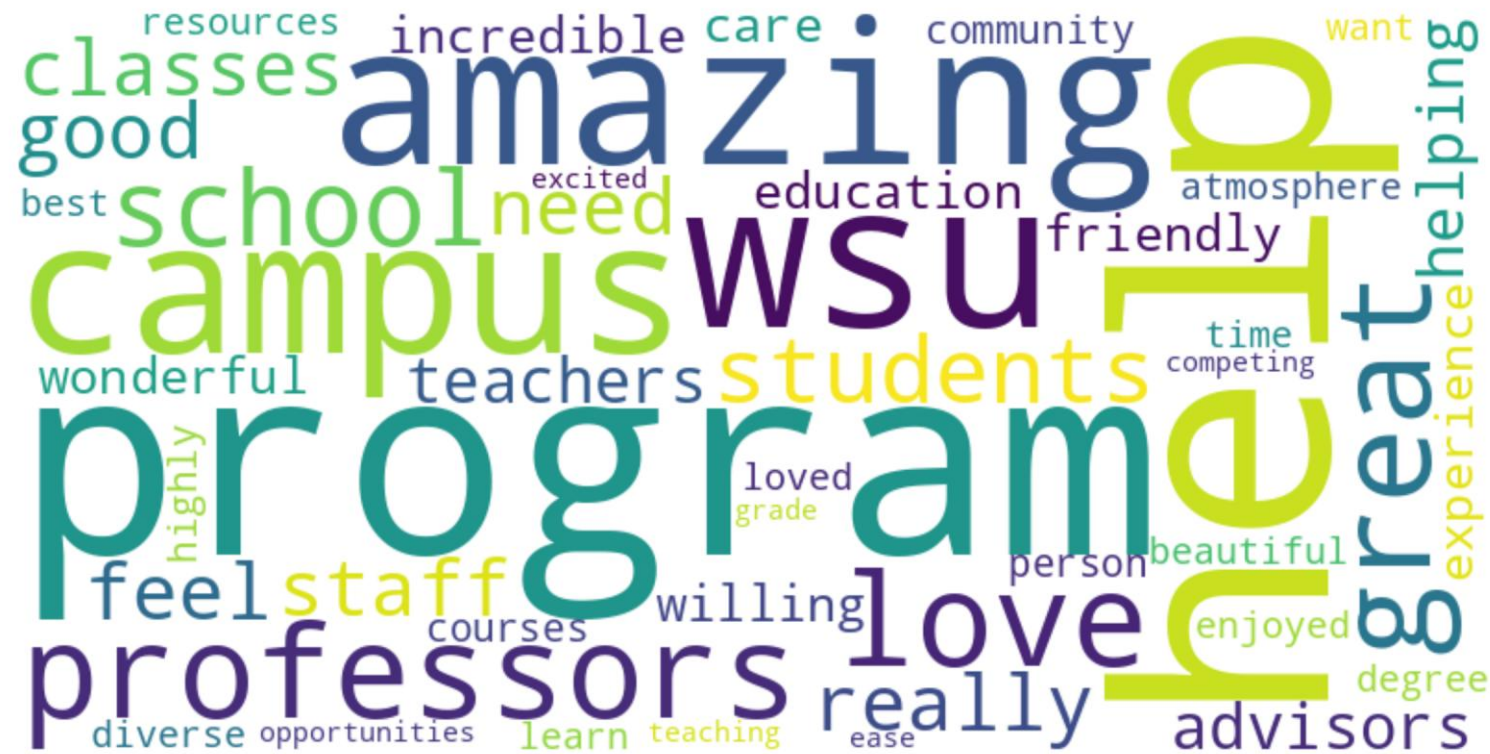


Singular Value Decomposition



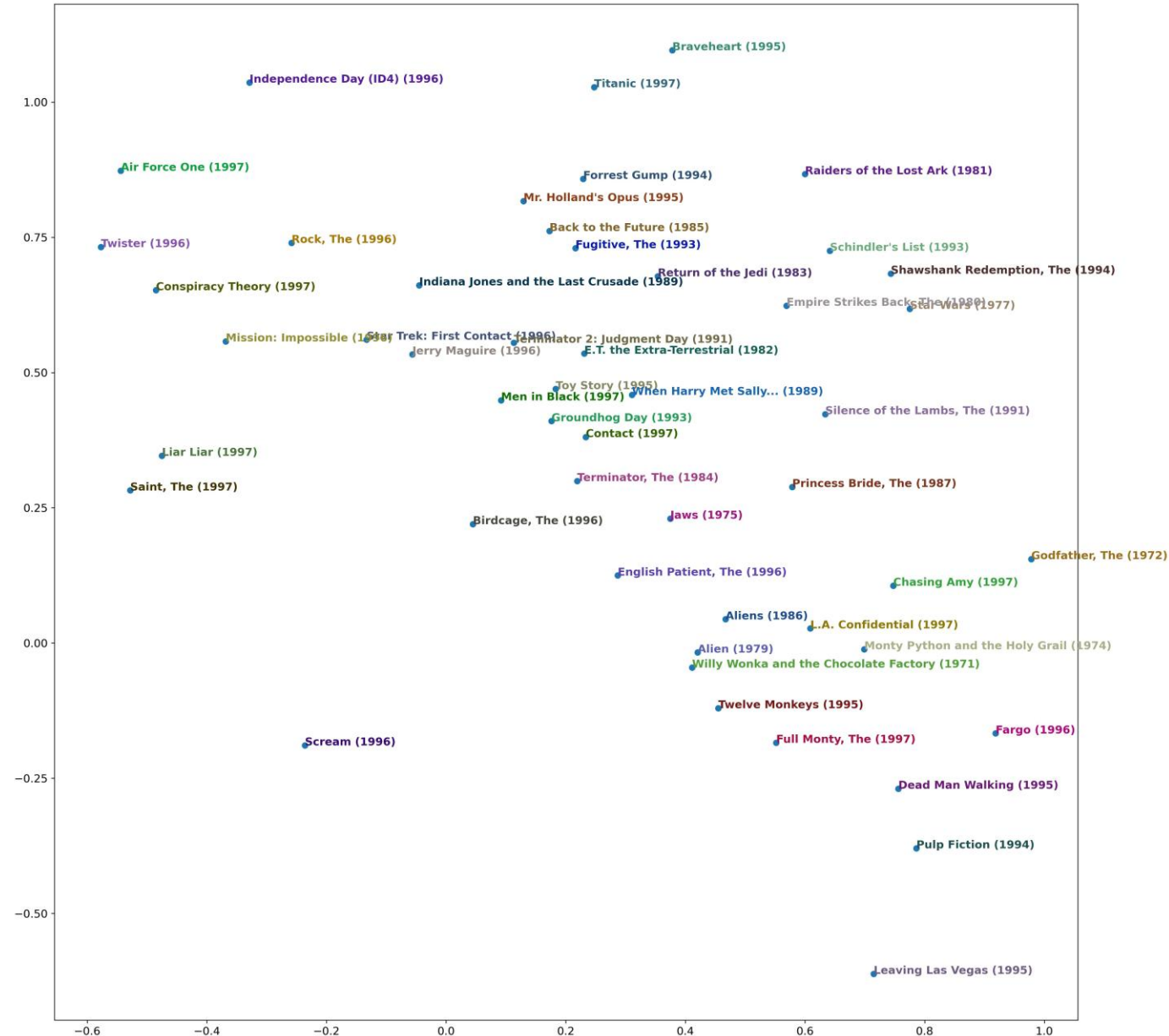
Applications

Topic Modeling with SVD



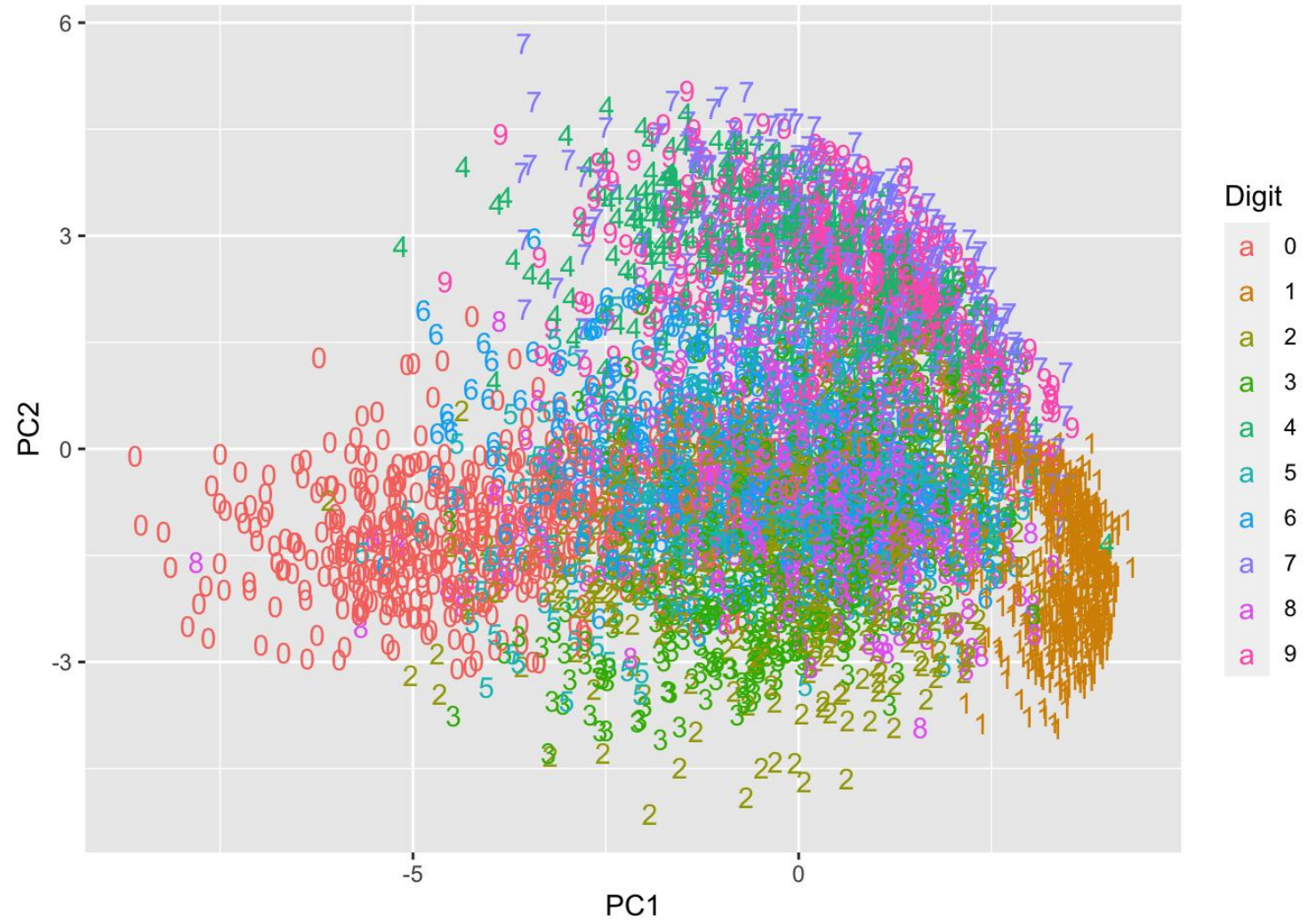
Applications

Movie Recommender Systems with SVD



Applications

Dimensionality reduction with SVD



Demo

